FRIDAY, SEPTEMBER 4TH, 2026

CROSS-DOMAIN SYNTHESIS

Larger Than Average

The world does not hand you a random sample of itself, and one formula says how far off it is.

3,086 words · 16-minute read

In 1922 a physicist at Tokyo Imperial University sat down to write about trams. Torahiko Terada had published on X-ray diffraction in 1913, at almost the same moment as the Braggs, and would found the university's Earthquake Research Institute in 1926; he also wrote essays for general readers, and the one he wrote that year concerned a complaint every commuter in the city would have recognised. The tram company published its intervals. Passengers, standing at the stop, waited longer than those intervals seemed to promise. Terada collected data on the Tokyo City trams and argued that the passengers were not imagining it and the company was not lying. An English translation of the essay, with a modern reanalysis of his figures, appeared in 2020, and its authors make the case that it is one of the earliest scientifically serious statements of what is now called the waiting-time paradox [1].

The paradox is worth stating cleanly, because the clean statement is what generalises. Suppose a service runs every ten minutes on average, and you arrive at the stop at a moment chosen without any reference to the timetable — determined by your day rather than by the trams. How long should you expect to wait?

The answer that presents itself is five minutes: half of ten, since you have landed at a random point inside a ten-minute gap. That answer is correct exactly once, in the case of a perfectly punctual service, and wrong in every other case. If the gaps are exponentially distributed — the case of arrivals with no memory at all, the Poisson process — your expected wait is not five minutes but ten. The full headway. Waiting for a bus that comes every ten minutes takes, on average, ten minutes.

The Interval You Land In

The reason is that you do not sample gaps. You sample instants, and instants are unevenly spread across gaps: a twenty-minute gap holds twice as many instants as a ten-minute one, so it is twice as likely to be the gap you land in. The timetable's gaps and the passenger's gaps are two different populations.

Write the intervals between successive arrivals as independent draws from a density f with mean μ and standard deviation σ. The interval covering an instant chosen uniformly along the timeline is not distributed as f but as its length-biased twin, in which each interval is reweighted by its own length:

$$\tilde{f}(x) = \frac{x\,f(x)}{\mu}.$$

Everything follows from that one line. The mean of the reweighted density is the second moment of the original divided by the first, which rearranges into a form worth memorising:

$$\mathbb{E}[\tilde{L}] \;=\; \frac{\mathbb{E}[L^2]}{\mu} \;=\; \mu\,(1+c^2), \qquad c = \frac{\sigma}{\mu}.$$

The inflation factor is one plus the squared coefficient of variation. And since, conditional on landing in an interval of length x, your arrival is uniform along it, the expected remaining wait is half of that:

$$\mathbb{E}[R] \;=\; \frac{\mathbb{E}[L^2]}{2\mu} \;=\; \frac{\mu}{2}\,(1+c^2).$$

This is the forward recurrence time of renewal theory, the subject of the sections Feller titled "Waiting Time Paradoxes" and, immediately after, "The Persistence of Bad Luck" [2]. Cox's 1962 monograph is the compact treatment [3], and the identification of the bus paradox with size-biased sampling as one phenomenon was made explicitly by Gupta in 1979 [4].

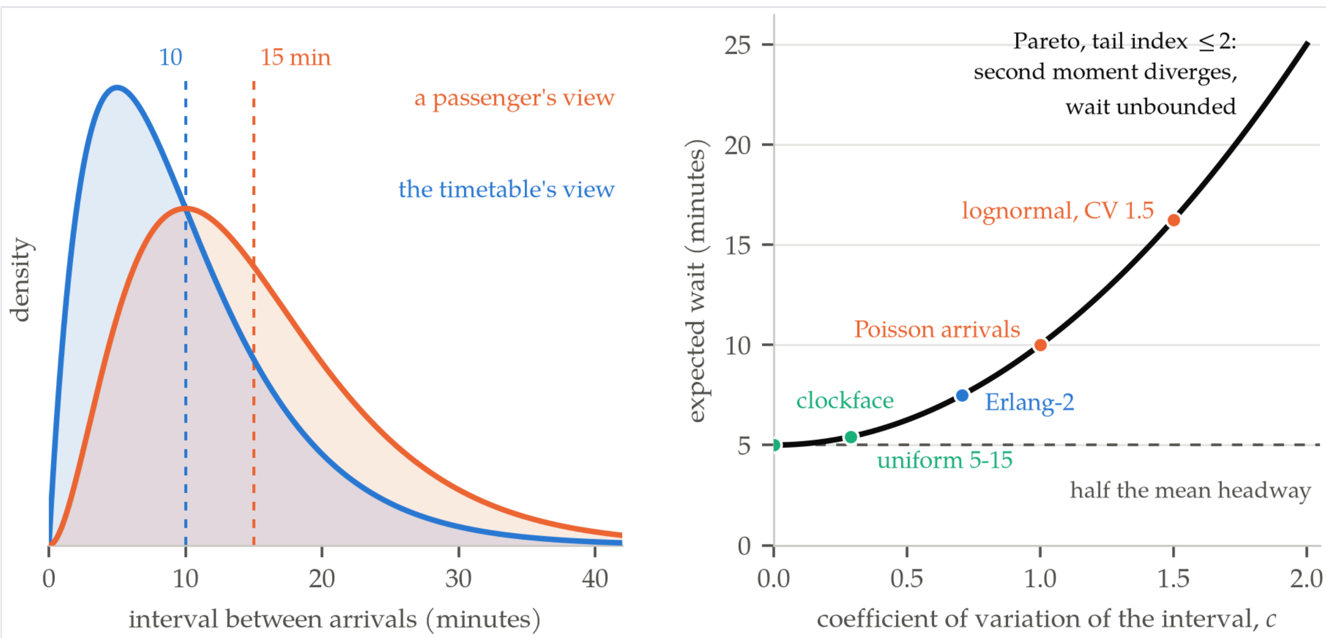

FIGURE The same arrival process seen two ways. Left: intervals between arrivals are Erlang-2 with a mean of ten minutes, and the interval a passenger actually lands in is the same family with its shape raised by one — a mean of fifteen minutes rather than ten. Right: the expected wait as a function of the coefficient of variation of the headway, for a mean headway of ten minutes. The dashed rule is the intuitive answer, correct only at the far left.

> **A SERVICE RUNS EVERY TEN MINUTES ON AVERAGE. CAN THE EXPECTED WAIT BE UNBOUNDED?**

It can, and the condition is exactly that the interval law have infinite variance. The formula depends on $\mathbb{E}[L^2]$ and not on the mean alone, which is a stronger requirement than the headway itself needs. For a Pareto interval law with tail index at or below 2, the mean headway is a perfectly ordinary ten minutes and the expected wait is infinite. At tail index 2.5 — still a well-behaved-looking distribution — the second moment is 180 and the expected wait is nine minutes, nearly twice the naive answer. The table below gives the finite cases, every value confirmed against a stationary simulation that probes two million instants along a long realisation of each process.

Interval law	c	Interval met (min)	Expected wait (min)
Clockface, exactly 10 min	0.000	10.00	5.00
Uniform, 5 to 15 min	0.289	10.83	5.42
Erlang-2	0.707	15.00	7.50
Pareto, tail index 2.5	0.894	18.00	9.00
Exponential (Poisson)	1.000	20.00	10.00
Lognormal, CV 1.5	1.500	32.50	16.25

The exponential row is the fixed point of the whole business. Any law with a coefficient of variation of one delivers an expected wait equal to the full mean headway, but the exponential is the only one for which the entire distribution of your remaining wait is identical to the distribution of a whole interval. Memorylessness is precisely the condition under which the time you have already stood there tells you nothing about how much longer you will, and recovering that characterisation of the exponential is what Gupta's paper does with the paradox [4]. Everything to the left of the exponential row is a service with some discipline; everything to the right is a service whose gaps are wilder than pure randomness, which describes rather a lot of real transport.

Larger Than Average, Everywhere

Nothing in that derivation is about time. Replace "interval" with any positive size and "instant" with any unit that is drawn in proportion to size, and the same factor appears.

The cleanest documented case is a school's own prospectus. In 1982 David Hemenway worked out the class sizes at the Harvard School of Public Health for the first quarter of the 1980–81 academic year: 111 courses, from a tutorial with a single student up to an epidemiology course with 229. The administration's average — total enrolments divided by number of courses — was 14.5. The average class size experienced by a student was over 78 [5].

Those two numbers are the same number viewed from opposite ends of the reweighting. A course is sampled by the registrar; a class is sampled by a student, and students, like instants, are distributed in proportion to size. Hemenway reported that only three courses exceeded 78 students, enrolling 105, 171 and 229, which is enough to reconstruct the arithmetic and check that both of his figures hold together. Total enrolments must be about 1,610, so the sum of squared class sizes must be about 125,700 for the student-weighted mean to reach 78.1 — implying a coefficient of variation of 2.09 across courses, since $14.5 \times (1 + 2.09^2) = 78.1$ exactly. The three named courses supply 31 per cent of the enrolments and 74 per cent of that second moment. Strip them out and the remaining 108 courses average 10.2 students with a student-experienced size of 29.9. Three rooms out of 111 do almost all the work, which is what a squared term does.

The same flip runs through any economy's firm statistics. Firms with fewer than twenty employees are 89 per cent of all American employer firms [6]; firms with fewer than five hundred, the official definition of small, employ 62.3 million people, or 45.9 per cent of private-sector workers [7]. Those two figures imply a private-sector workforce of about 135.7 million, the right order for the period, and together they make the point: almost every firm is small, and most jobs are not at one. A random firm and a random job are questions about different distributions.

Two older examples show the bias arising from physical apparatus rather than from a definition. Cox, writing in 1969 on sampling in industry, pointed out that the standard way of estimating mean fibre length in a textile — gripping a tuft with a pincer and measuring what is caught — systematically misses short fibres, because a fibre's chance of being gripped scales with its length [8]. And Wicksell's 1925 corpuscle problem, the founding paper of stereology, concerns a solid full of spheres cut into thin sections: a large sphere is likelier to meet the cutting plane than a small one, so the observed circles are biased twice over, in which sphere they represent and in how large a circle it shows [9].

The Speedometer and the Queue

Traffic engineering had to confront the same arithmetic to make its instruments trustworthy. A loop detector buried in a lane measures the speed of every vehicle that crosses it and reports the average. That average is not the quantity that determines how long a journey takes.

The reason is the familiar one: a fast vehicle covers more ground per unit time, so it crosses more detectors per unit of its presence on the road, and a point sensor therefore samples vehicles in proportion to their speed. The average it reports — the time-mean speed — is the size-biased mean of the speeds actually present in space, and Wardrop set the relation out in his 1952 paper to the Institution of Civil Engineers:

$$\bar{v}_t \;=\; \bar{v}_s \;+\; \frac{\sigma_s^2}{\bar{v}_s},$$

where $\bar{v}_s$ is the space-mean speed and σ_s its standard deviation across vehicles [10]. It is $\mu(1+c^2)$ again, written for speeds. On a simulated lane with a space-mean speed of 60 kilometres an hour and a standard deviation of 20, a point detector reports 66.6 — an overstatement of 11 per cent, produced by no instrument error whatsoever. Travel time is governed by the space mean, so a naive detector average flatters the road.

Queueing theory met the bias from the other direction. A customer arriving at a single server with a queue in progress finds the server part-way through a job, and the expected time until that job finishes is not half the mean service time but $\mathbb{E}[S^2]/2\mathbb{E}[S]$ — the same residual-life expression, with service times in place of headways. This is why the mean wait in a queue with general service times, derived by Pollaczek in 1930 and recast probabilistically by Khinchine two years later, carries a second moment [11], [12]:

$$W_q = \frac{\lambda\,\mathbb{E}[S^2]}{2(1-\rho)} = \frac{\rho\,\mathbb{E}[S]\,(1+c_S^2)}{2(1-\rho)}.$$

Hold the arrival rate and the mean service time fixed, so that utilisation stays at 80 per cent, and vary only the variability of the service times. Deterministic service gives a mean wait of two service times, exponential service four, and service with a coefficient of variation of 2 gives ten. A fivefold difference in delay, with the server doing the same work per job on average. Variance is not a second-order correction to congestion; it is most of congestion.

Which closes a loop back to the tram stop, because transit headways do not merely happen to be variable. Newell and Potts showed in 1964 that a bus route is dynamically unstable: a bus running slightly late finds more passengers waiting, dwells longer and falls further behind, while the bus behind it catches up into an emptier stop and runs early [13]. The system manufactures bunching, and bunching manufactures exactly the variance the second moment punishes.

When the Bias Changes the Conclusion

So far the bias has cost accuracy. In medicine it has cost conclusions.

Suppose you want to know how long people survive after the onset of dementia. The obvious design is to find people who have dementia, follow them, and record when they die. The trouble is that a study which enrols people who already have the disease can only enrol people whose disease has lasted long enough for the study to find them. Rapidly fatal cases are systematically absent — not under-represented but absent, in proportion to how short they were. It is Cox's pincer, gripping patients instead of fibres.

Estimates of median survival after onset had ranged from five to over nine years. In 2001 Christina Wolfson and colleagues went back to the Canadian Study of Health and Aging, a prevalent cohort of 821 people with dementia, and estimated survival twice: once naively and once with a correction for length bias. The unadjusted median was 6.60 years. The adjusted median was 3.30 years, with a 95 per cent confidence interval of 2.7 to 4.0 [14]. The literature had been overstating survival by a factor of two, and the reason was not measurement, funding or follow-up. It was that the sample had been drawn in proportion to duration.

The factor of two is worth pressing on. If the true durations were exponential, pure length bias inflates the mean by exactly $1 + c^2 = 2$ and the median by 2.42, since the length-biased version of an exponential is a gamma with shape 2, whose median is 1.678 times the scale against the exponential's 0.693. The observed inflation of 2.00 sits just inside what pure length bias predicts for the simplest duration law — not a derivation of the result, but a sign that the mechanism identified is large enough to be the whole story.

The same structure had been formalised for screening three decades earlier. Zelen and Feinleib's 1969 paper set out the stable-disease model in which a case spends a period in a detectable pre-clinical state before becoming clinically apparent [15]. A screening programme intersects that period the way Wicksell's plane intersects a sphere, and so preferentially catches the slow cases — precisely the ones with the best prognosis. Screen-detected cancers will appear to do better than symptom-detected ones even if screening does nothing at all, which is why screening trials are designed around mortality in the whole invited population rather than survival among the detected.

Your Friends Are Not a Random Sample

The most-cited instance of the formula was published in sociology. In 1991 Scott Feld asked why people so consistently feel less popular than their friends and answered that they are usually right, for reasons having nothing to do with them. Sample a person and you get a degree drawn from the degree distribution; sample a friendship and follow it, and you arrive at a person in proportion to how many friendships they have [16]:

$$\mathbb{E}[\text{degree of a random neighbour}] \;=\; \frac{\mathbb{E}[k^2]}{\mathbb{E}[k]} \;=\; \bar{k} + \frac{\sigma_k^2}{\bar{k}}.$$

The gap is the variance of the degree distribution over its mean, and in real social networks that variance is enormous. The inequality is strict for any network in which people differ in how many friends they have, which is every network, and it needs no assumption about who befriends whom.

Two research programmes have turned the bias into an instrument, which is the most satisfying thing that can happen to a bias. Early warning of an epidemic means monitoring unusually well-connected people, and finding them normally requires knowing the whole network — but sampling friends of randomly chosen people delivers high-degree people for free, which is how Garcia-Herranz and colleagues built a sentinel group whose infection curve led the population's by days and sometimes considerably more [17]. The public-health version is older: Cohen, Havlin and ben-Avraham's acquaintance immunisation asks each of a random set of people to name a contact and vaccinates the contact, concentrating doses on the well-connected while requiring no map of the network at all [18].

The Harmonic Correction

The bias is one formula, and so, remarkably, is the fix.

Take the length-biased density and compute the expectation not of x but of $1/x$:

$$\mathbb{E}\left[\frac{1}{\tilde{L}}\right] \;=\; \int_0^\infty \frac{1}{x} \cdot \frac{x f(x)}{\mu}\, dx \;=\; \frac{1}{\mu}.$$

The reciprocal of the mean reciprocal of a size-biased sample is the mean of the original population, exactly, for every distribution with a finite mean. The estimator, in other words, is the harmonic mean of the observations, n divided by the sum of their reciprocals. So a passenger who records the gaps she happens to land in and averages them overstates the headway by $1 + c^2$, while one who averages the reciprocals and inverts recovers the timetable's own mean; simulation confirms it to four digits across gamma, lognormal and uniform interval laws. It is also why traffic engineering's standing instruction is that space-mean speed is the harmonic mean of spot speeds — not a convention but the exact inverse of Wardrop's relation.

The general form of the correction predates every application above. Horvitz and Thompson established in 1952 that if unit i enters a sample with known probability π_i, weighting each observation by $1/\pi_i$ gives an unbiased estimate of the population total [19]. Length bias is the special case $\pi_i \propto x_i$, and the harmonic mean is what inverse-probability weighting reduces to there. The distributional theory has its own lineage: Fisher's 1934 paper on methods of ascertainment, written about human genetics, where families reach a geneticist's attention through their affected members [20]; Rao's 1965 generalisation to arbitrary weight functions [21]; and Patil and Rao's 1978 treatment, which works through wildlife populations and human family sizes side by side and made "size-biased" standard usage [22].

IF THE CORRECTION IS ONE LINE, WHY DOES THE BIAS KEEP BEING REDISCOVERED?

Partly because the weight is often invisible. Nobody at the Harvard School of Public Health was doing arithmetic wrong; both averages were correct answers to different questions, and only one of them appeared in the prospectus. But there is a harder reason, and it is the honest limit of the apparatus: inverse-probability weighting can only reweight units that appear in the sample. Units of size zero have no chance of being drawn, and no weighting recovers them. The course nobody enrolled in is absent from the student-weighted average, the fibre too short for the pincer is invisible to the pincer, and the tram interval during which the service was suspended contains no passengers to report it. Reweighting fixes a distortion in what you saw; it cannot tell you what could not have been seen.

What is left is a discipline rather than a formula. Before averaging anything, ask what did the sampling — an instant, a student, a job, a passing vehicle, a friendship, a cutting plane — and whether that unit is drawn in proportion to the thing being measured. If it is, the answer is not wrong by a little. It is wrong by one plus the squared coefficient of variation, and in the cases that matter that number is nowhere near one.

REFERENCES

[1] N. Masuda and T. Hiraoka, "Waiting-Time Paradox in 1922," *Northeast Journal of Complex Systems*, vol. 2, no. 1, art. 1, 2020.

[2] W. Feller, *An Introduction to Probability Theory and Its Applications*, vol. II, 2nd ed. New York: Wiley, 1971, ch. I, sec. 4-5.

[3] D. R. Cox, *Renewal Theory*. London: Methuen, 1962.

[4] R. C. Gupta, "Waiting Time Paradox and Size Biased Sampling," *Communications in Statistics - Theory and Methods*, vol. 8, no. 6, pp. 601-607, 1979.

[5] D. Hemenway, “Why Your Classes Are Larger than ‘Average’,” *Mathematics Magazine*, vol. 55, no. 3, pp. 162-164, 1982.

[6] US Census Bureau, *Statistics of U.S. Businesses: 2022 Annual Data Tables*, 2025.

[7] US SBA Office of Advocacy, *Frequently Asked Questions About Small Business*, 2026.

[8] D. R. Cox, “Some Sampling Problems in Technology,” in *New Developments in Survey Sampling*, N. L. Johnson and H. Smith, Eds. New York: Wiley-Interscience, 1969, pp. 506-527.

[9] S. D. Wicksell, “The Corpuscle Problem. A Mathematical Study of a Biometric Problem,” *Biometrika*, vol. 17, pp. 84-99, 1925.

[10] J. G. Wardrop, “Some Theoretical Aspects of Road Traffic Research,” *Proceedings of the Institution of Civil Engineers*, vol. 1, no. 3, pp. 325-362, 1952.

[11] F. Pollaczek, “Über eine Aufgabe der Wahrscheinlichkeitstheorie. I,” *Mathematische Zeitschrift*, vol. 32, pp. 64-100, 1930.

[12] A. Y. Khinchine, “Mathematical Theory of a Stationary Queue,” *Matematicheskii Sbornik*, vol. 39, no. 4, pp. 73-84, 1932.

[13] G. F. Newell and R. B. Potts, “Maintaining a Bus Schedule,” in *Proc. 2nd Australian Road Research Board Conf.*, vol. 2, pp. 388-393, 1964.

[14] C. Wolfson *et al.*, “A Reevaluation of the Duration of Survival after the Onset of Dementia,” *New England Journal of Medicine*, vol. 344, no. 15, pp. 1111-1116, 2001.

[15] M. Zelen and M. Feinleib, “On the Theory of Screening for Chronic Diseases,” *Biometrika*, vol. 56, no. 3, pp. 601-614, 1969.

[16] S. L. Feld, “Why Your Friends Have More Friends Than You Do,” *American Journal of Sociology*, vol. 96, no. 6, pp. 1464-1477, 1991.

[17] M. Garcia-Herranz *et al.*, “Using Friends as Sensors to Detect Global-Scale Contagious Outbreaks,” *PLoS ONE*, vol. 9, no. 4, art. e92413, 2014.

[18] R. Cohen, S. Havlin and D. ben-Avraham, “Efficient Immunization Strategies for Computer Networks and Populations,” *Physical Review Letters*, vol. 91, no. 24, art. 247901, 2003.

[19] D. G. Horvitz and D. J. Thompson, “A Generalization of Sampling Without Replacement from a Finite Universe,” *Journal of the American Statistical Association*, vol. 47, pp. 663-685, 1952.

[20] R. A. Fisher, “The Effect of Methods of Ascertainment upon the Estimation of Frequencies,” *Annals of Eugenics*, vol. 6, pp. 13-25, 1934.

[21] C. R. Rao, “On Discrete Distributions Arising out of Methods of Ascertainment,” *Sankhyā Series A*, vol. 27, pp. 311-324, 1965.

[22] G. P. Patil and C. R. Rao, “Weighted Distributions and Size-Biased Sampling with Applications to Wildlife Populations and Human Families,” *Biometrics*, vol. 34, pp. 179-189, 1978.

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